PARISHRAM 2.0

CLASS 12TH 2026

Relations and Functions

MATHS **LECTURE-5**

BY-MANDAR SIR

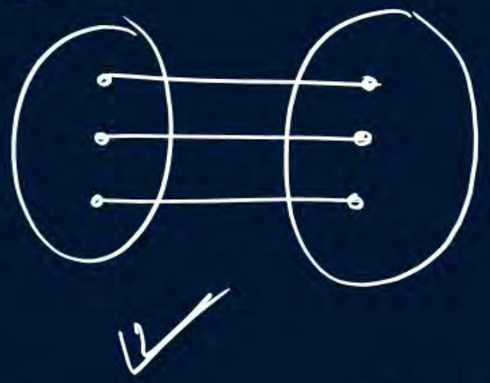


TOPICS TO BE COVERED

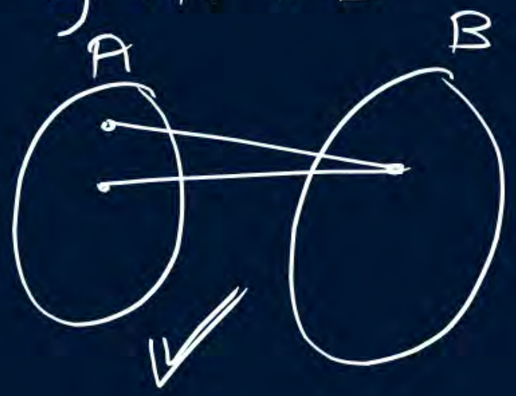
1. Types of Functions
2. Questions Related to Types

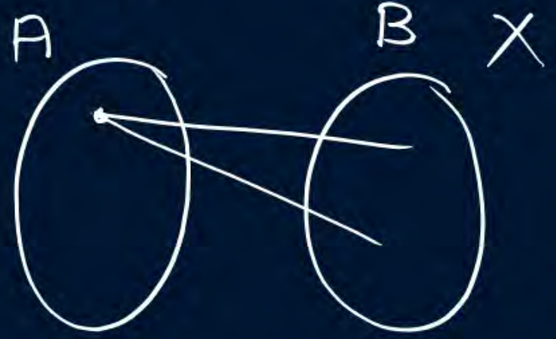
Functions [Conditions]

$f: A \to B$

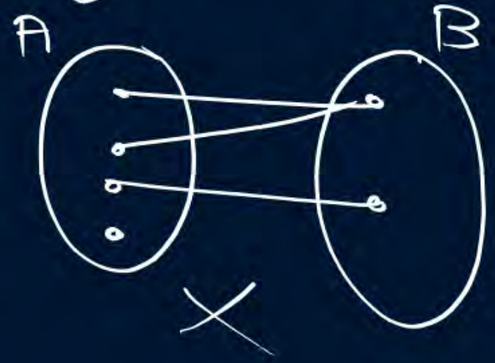


$f: A \to B$





$f: A \to B$



Types of Functions

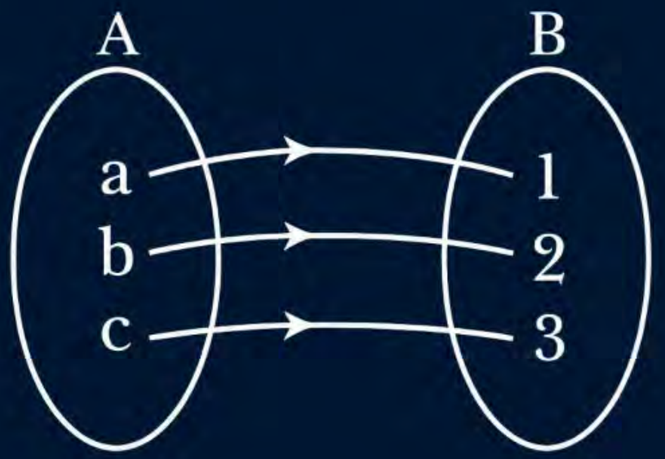
1. One-One Function (Injective)
2. Many-One Function
3. Onto Function (Surjective)
4. Bijective Function (one-one & onto)
5. Into Function

One-One (Injective) Function

(Ek connected hoga ek se hi)

A function $f : A \to B$ is called a one-one or injective function, if distinct elements of A have distinct images in B, i.e. for every $a_1, a_2 \in A$ if $a_1 \neq a_2$, then $f(a_1) \neq f(a_2)$ and if $f(a_1) = f(a_2)$ then $a_1 = a_2$.

e.g. Let $f : A \to B$ be a function represented by the following diagram.

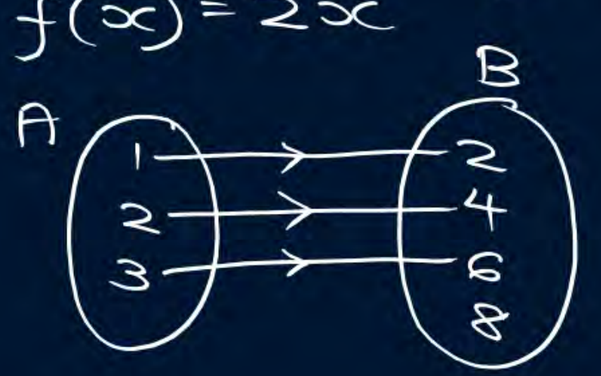


$f : A \to B$

$A = \{1, 2, 3\}$

$B = \{2, 4, 6, 8\}$

$f(x) = 2x$



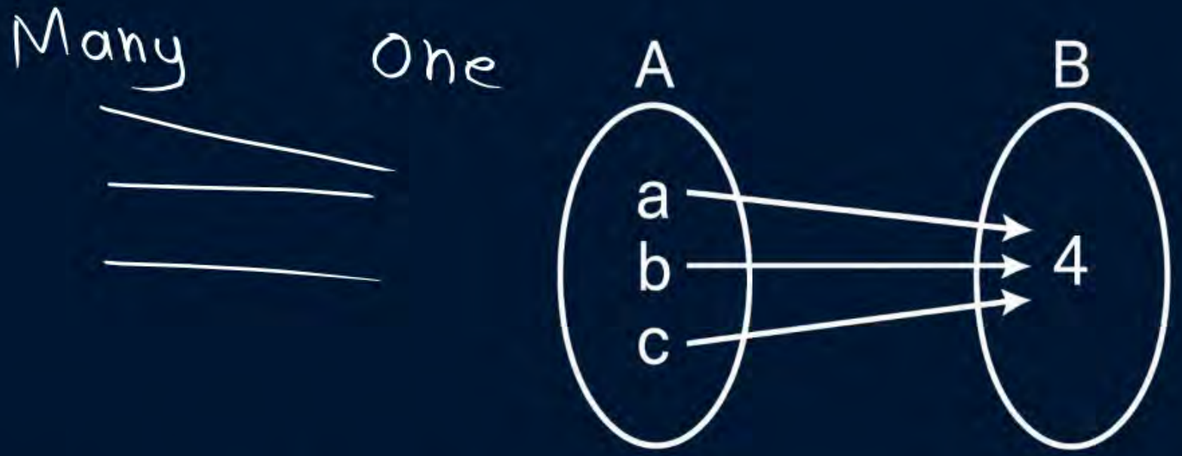
$f(1) = 2(1) = 2$

Here, f is a one-one function, because each element have distinct image.

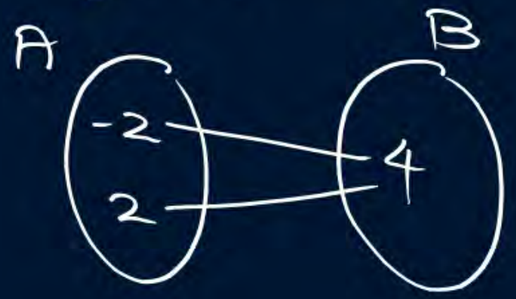
Many-One Function

A function $f : A \to B$ is called a many-one function, if there exist atleast two distinct elements in A, whose images are same in B, i.e. if there exist $a_1, a_2 \in A$ such that $a_1 \neq a_2$ and $f(a_1) = f(a_2)$, then f is many-one. In other words, a function $f : A \to B$ is called a many-one function, if it is not one-one.

e.g. Let $f : A \to B$ be a function represented by the following diagram.



$f(x) = x^2$



$f(-2) = (-2)^2 = 4$

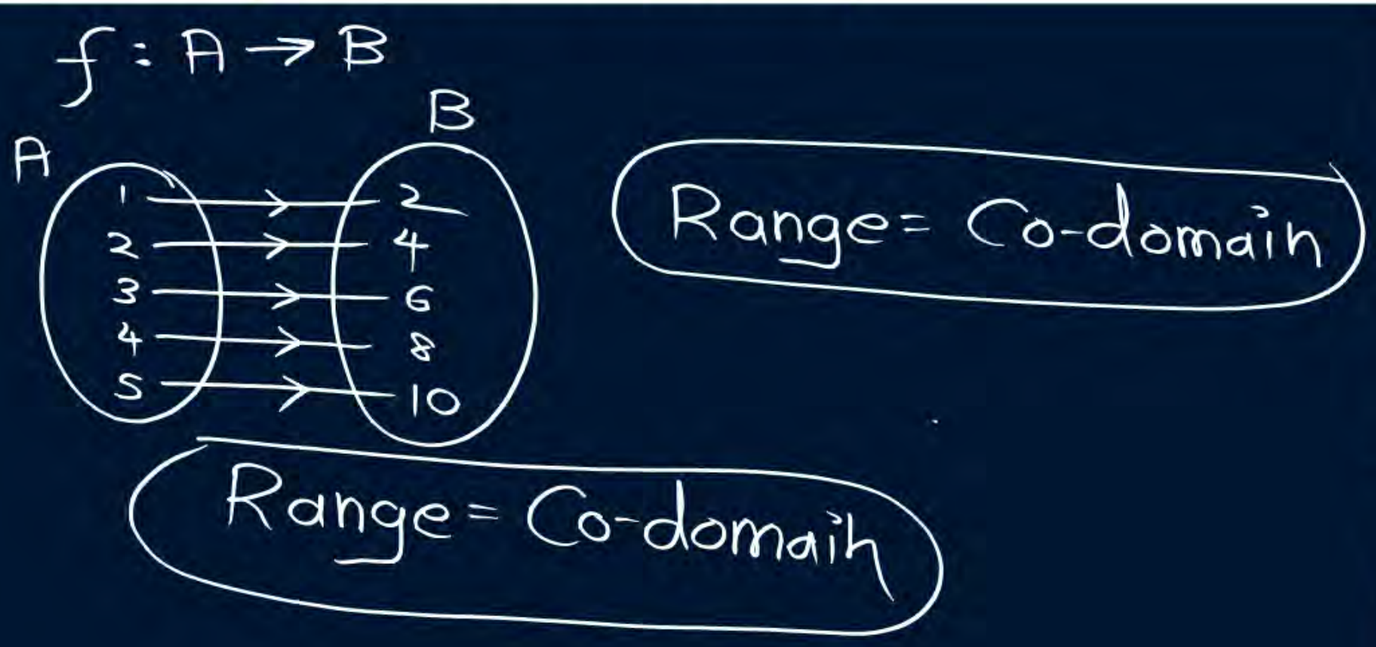
$f(2) = 2^2 = 4$

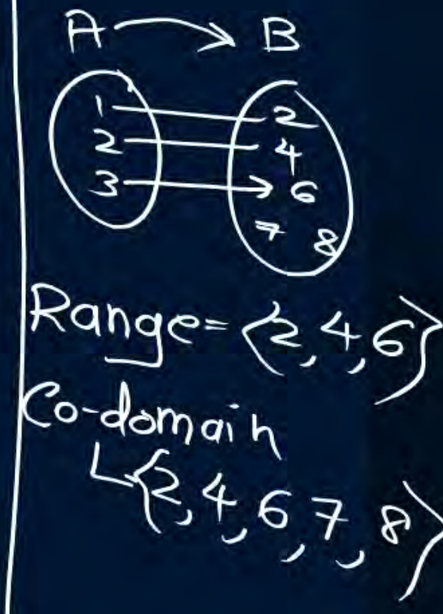
Here, f is many one function, because a_1 and a_3 have same image b_2.

Onto Function

A function f from set A to set B is called onto function if each element of set B has a preimage in set A or range of function f is equal to the codomain i.e., set B.

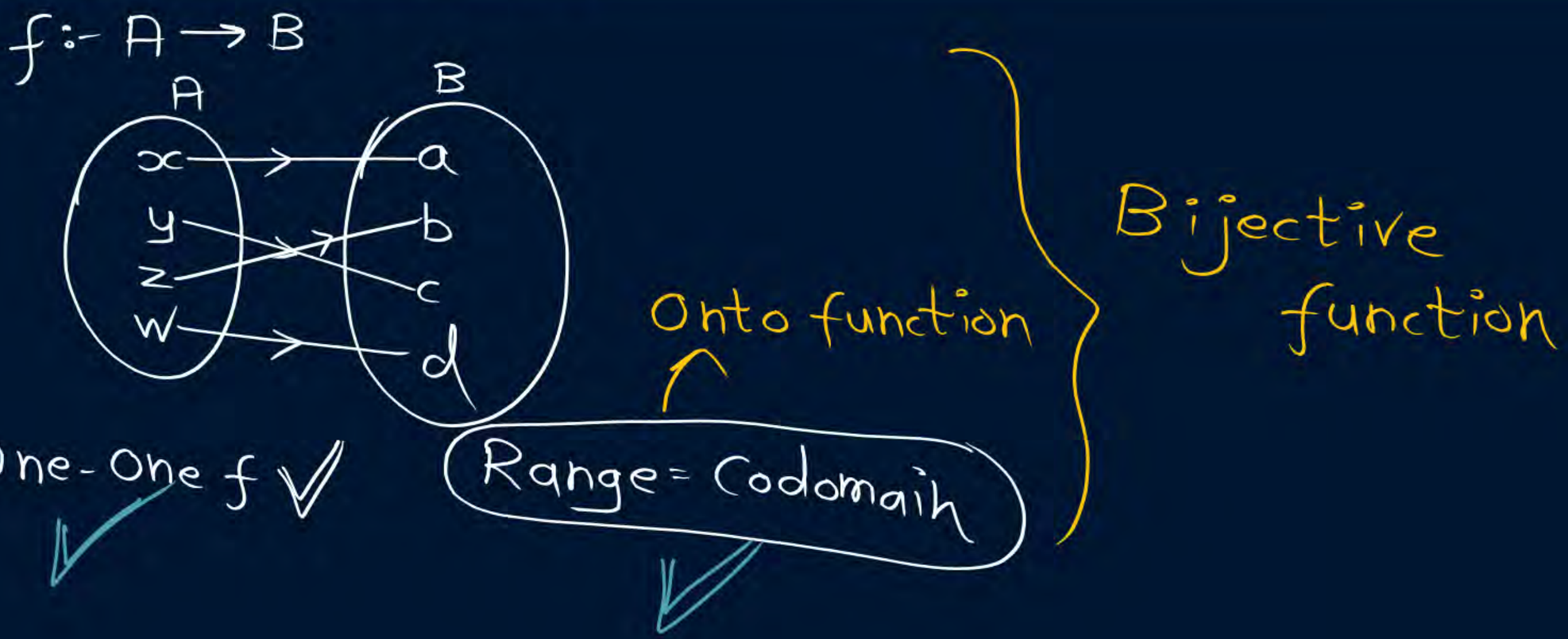
Onto function is also called surjective function.





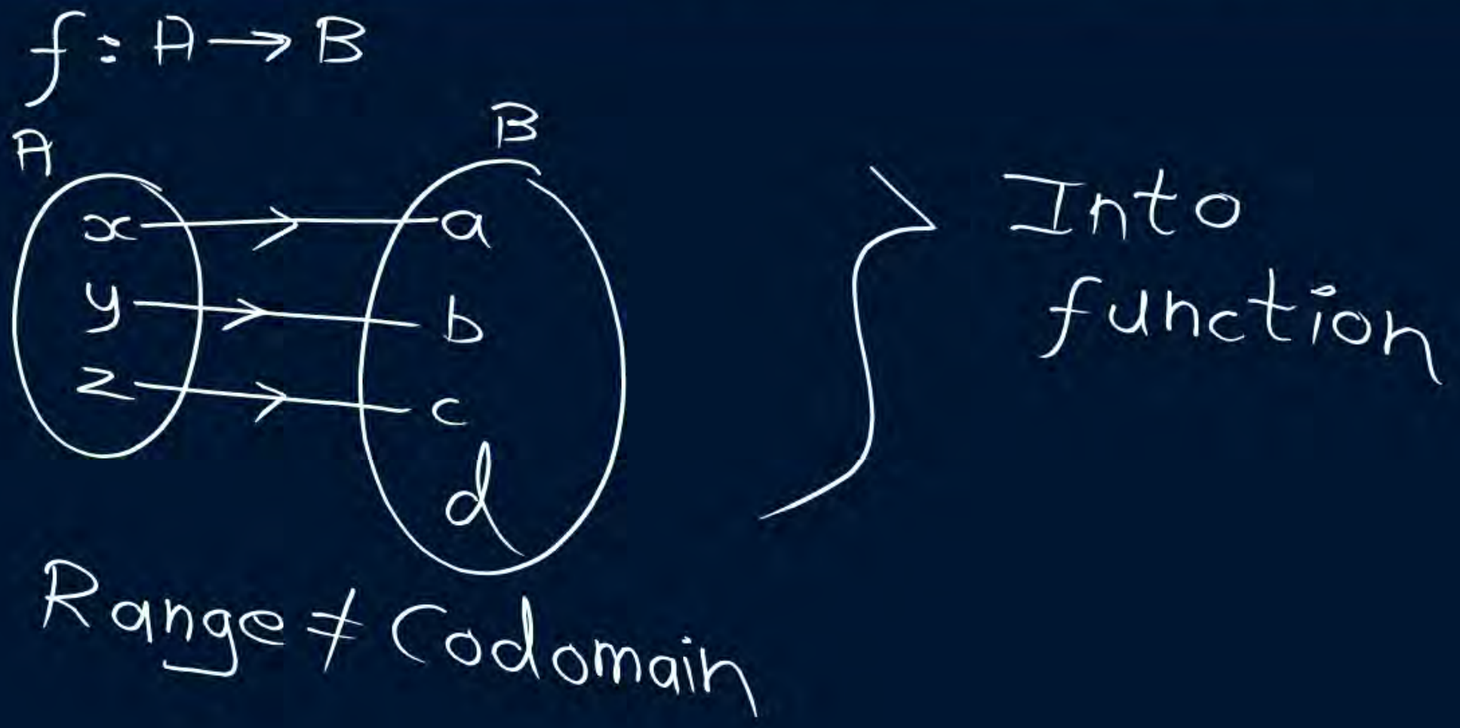
Bijective Function

A function f from set A to set B is called bijective function if it is both one-one function and onto function.



Into Function

A function $f : A \to B$ is an into function if there exists atleast one element in B having no pre-image in A.
In other words, $f : A \to B$ is an into function if it is not an onto functions.



To Check one-one and onto

One-One

$x_1\ x_2 \in R$

$f(x_1) = f(x_2)$

$x_1 = x_2$ ✓

Onto

$y = f(x)$

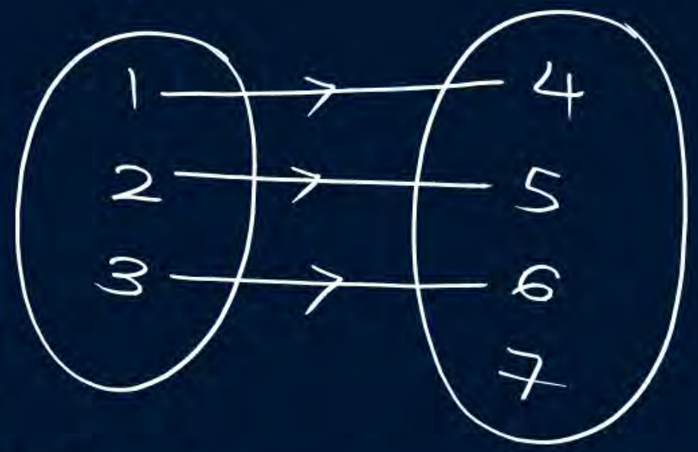
For every y there is x in domain.

Agar → apka y aya
↳ x [Domain]

Question

Let $A = \{1, 2, 3\}$, $B = \{4, 5, 6, 7\}$ and let $f = \{(1, 4), (2, 5), (3, 6)\}$ be a function form A to B. Show that f is one-one.

$f: A \to B$



Into f^r

Range $\neq$ Co-domain

$f: A \to B$

One-one function

Ek element $\longrightarrow$ Ek se connected h

Question

Show that the function $f : R^* \to R^*$ defined by $f(x) = 1/x$ is one-one and onto, where R is the set of all non-zero real numbers. Is the result true, if the domain R^* is replaced by N with co-domain being same as R^* ?

$f(x) = \frac{1}{x}$

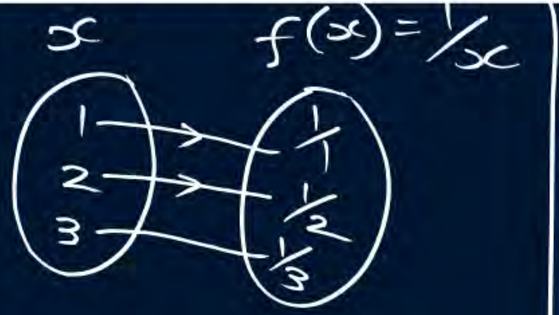
For one-one fn

$f(x_1) = f(x_2)$

$\frac{1}{x_1} = \frac{1}{x_2}$

$x_2 = x_1$

One-one fn ✓



Onto function

$f(x) = \frac{1}{x}$

$y = \frac{1}{x}$

$x = \frac{1}{y}$

Onto function

$y \to$ Real value put

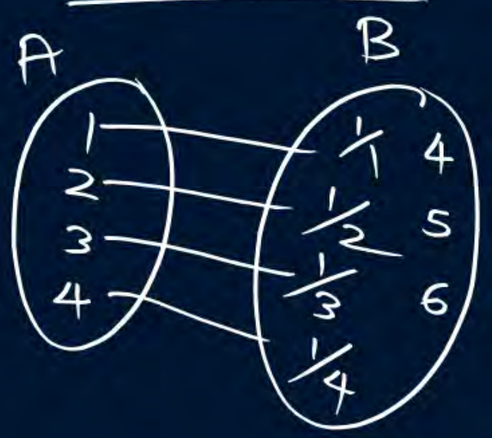
$x \to$ Real value ✓

$f: R \to R$

Now ↓ replace

$f: N \to R$ $\quad f(x) = \frac{1}{5x}$

One-one



Range ≠ Co-domain

One-One f^n ✓

$f(1) = \frac{1}{1}$

$f(2) = \frac{1}{2}$

$f(3) = \frac{1}{3}$

$f(4) = \frac{1}{4}$

Onto function

$f(x) = \frac{1}{x}$

$y = \frac{1}{x}$

$x = \frac{1}{y}$

$x = \frac{1}{\sqrt{5}}$ $\quad y = \sqrt{5}$

$x \neq \frac{1}{\sqrt{5}}$ (Not my onto f^n)

Question

One-one

Check the injectivity and surjectivity of the following functions:

(i) $f : N \to N$ **given by** $f(x) = x^2$

$f(x) = x^2$

$f : N \to N$

One
One ✓

For one-one

$f(x_1) = f(x_2)$

$x_1^2 = x_2^2$

$x_1^2 - x_2^2 = 0$

$(x_1 + x_2)(x_1 - x_2) = 0$

$x_1 = -x_2$ ✗

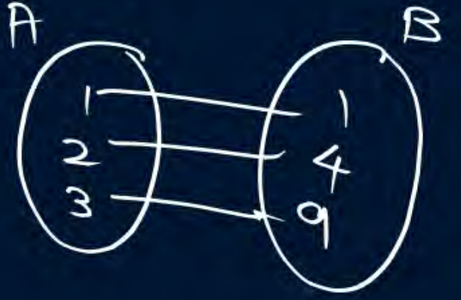
$x_1 = x_2$ ✓

$f(x) = x^2$

$f(1) = 1^2 = 1$

$f(2) = 2^2 = 4$

$f(3) = 3^2 = 9$



Onto function

$f(x) = x^2$

$y = x^2$

$\sqrt{y} = x$

If $y = 7$

$\sqrt{7} = x$

Onto fn ✗

Question

Check the injectivity and surjectivity of the following functions:

(ii) $f : Z \to Z$ given by $f(x) = x^2$

$Z \to$ integers

One-one

$f(x_1) = f(x_2)$

$x_1^2 = x_2^2$

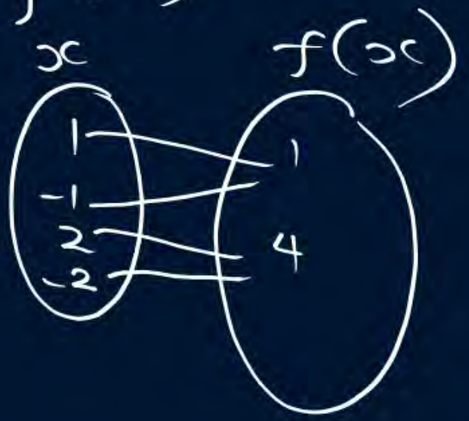
$x_1^2 - x_2^2 = 0$

$(x_1 + x_2)(x_1 - x_2) = 0$

$x_1 = -x_2$ ✓ $\quad x_1 = x_2$ ✓

Not a one-one function

$f(x) = x^2$



Onto function

$f(x) = x^2$

$y = x^2$

$\sqrt{y} = x$

$y = -11$

$\sqrt{-11} = x$ ✗

Onto function ✗

Question

Check the injectivity and surjectivity of the following functions:

(iii) $f : R \to R$ given by $f(x) = x^2$

One-One function

$f(x_1) = f(x_2)$

$x_1^2 = x_2^2$

$x_1^2 - x_2^2 = 0$

$(x_1 + x_2)(x_1 - x_2) = 0$

$x_1 = -x_2 \qquad x_1 = x_2$

$x_1 = \pm x_2$

✗

Onto function

$f(x) = x^2$

$y = x^2$

$\sqrt{y} = x$

$y = -3$

$\sqrt{-3} = x$

$\sqrt{-3}$ → Real

→ Imaginary

Onto → ✓

→ ✗

Homework

$f : N \to N$

$f(x) = x^3$

Check injective & surjective

THANK YOU